

Tony Waiguru

Python Backend Developer

Real-time systems, fintech and automation

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Python backend developer with a telecommunications engineering background and independent projects spanning real-time event processing, account orchestration, transaction reconciliation, and trading software. Built Python/Rust services and a TypeScript trading workspace, with emphasis on concurrency, durable state, API integration, and failure recovery. Earlier roles at KBC, KCAA, and Heaven Bound TV developed practical troubleshooting experience across broadcast, aviation communications, and IT networks.

Technical skills

Backend and systems: Python, asyncio, Rust, Tokio, SQL, SQLite WAL, aiohttp, Pydantic, WebSockets, STOMP.

Product and integrations: TypeScript, React, Next.js, Vite, Tailwind CSS, Deriv API, OAuth, Telegram, Patchright, MQL5.

Operations and quality: Linux, AWS EC2, systemd, Cloudflare Workers/D1, Drizzle, Git, Bash, PowerShell, pytest, Mypy, Ruff, Clippy, Prometheus.

Selected software projects

Independent project development | Selected work documented in 2026

Talon | Real-time orchestration and durable transaction workflows

Python / Rust / Tokio / STOMP and WebSockets / SQLite / systemd

- Separated latency-sensitive Rust event workers from Python account actors, a dedicated State service, and a Telegram operator console; kept browser and database work outside the networking critical path.
- Implemented bounded queues, event deduplication, independent account writer lanes, and account-scoped recovery to limit contention and isolate failures.
- Built durable transaction workflows with explicit acknowledgement, settlement, and reconciliation states; ambiguous outcomes block resubmission. Centralized SQLite WAL writes, migrations, backups, and archive operations.
- Documented release validation included 393 Python and 41 Rust tests, strict static checks, and six configured accounts reaching readiness in the 24 August 2026 release record.

VolaPilot | Trading workspace and broker API integration

TypeScript / React / Next.js / Cloudflare Workers and D1 / Deriv API

- Built a workspace combining live market data, entry analysis, Manual/Bulk/Auto order flows, practice, replay, and trade review, using a bounded 1,000-tick history per market.
- Integrated authentication and account lifecycle handling with a shared execution lock, risk controls, encrypted sessions, and explicit recovery for interrupted or uncertain purchases.
- Corrected stale entry evidence after feed interruptions and reduced repeated selected-rule calculation. A September 2026 local benchmark measured 2.20x faster Over and 3.01x faster Under refreshes with equivalent continuous-feed results.
- Added operational records, an owner console, saved strategy profiles, and regression coverage for scanner behavior, replay, account recovery, execution paths, and request security.

Performance figures describe local model calculation time. They do not represent broker latency or trading returns.

Further software projects

Aegis Meridian | Research, MT5 execution and audit telemetry

Python / MQL5 / Rust / SQLite / JSON Schema / Prometheus

- Implemented an event-driven research engine with cost-aware simulation, broker-aware sizing, walk-forward evaluation, bootstrap uncertainty, and structured release verdicts.
- Built a guarded MQL5 Expert Advisor and a read-only Rust telemetry sidecar with validated journals, a SHA-256 audit chain, health endpoints, and metrics.
- Aligned components through canonical configuration, schemas, and golden vectors. The August 2026 record reports 16 Python and five Rust tests passing, zero-error MQL5 compilation, and rejection of an edge-free research candidate. The project remains a research/shadow system.

Tyche | Python concurrency and runtime reliability

Python / asyncio and uvloop / websockets / Patchright / Linux

- Developed supervised per-site event workers, bounded queues, batched persistence, crash-safe JSON replacement, controlled logging, and a legacy-engine rollback path.
- Added event-loop lag, dispatch-to-wire timing, thread/file-descriptor counts, memory, and queue-pressure diagnostics; corrected a supervisor loop that could continuously consume one CPU core.

Earlier projects: Helios and Spectre developed session automation, WebSocket integration, strategy persistence, withdrawal workflows, and Telegram monitoring/control.

Professional experience

Telecommunications Technician | Kenya Broadcasting Corporation (KBC)

Nairobi | Jan 2024 - Apr 2024

- Maintained and troubleshooted radio broadcasting equipment and transmission systems spanning satellite, microwave, and fibre links; worked with colleagues to resolve technical faults and support broadcast operations.

Aeronautical Telecommunications Officer | Kenya Civil Aviation Authority (KCAA)

Mombasa | Feb 2022 - Apr 2022

- Worked with radio, radar, navigation aids, air traffic control equipment, and networking; developed practical fault-finding skills across aeronautical telecommunications systems.

Broadcast Technician | Heaven Bound TV

Githurai | Aug 2021 - Nov 2021

- Supported broadcast equipment and transmission channels alongside network administration and user support, combining telecommunications and IT troubleshooting.

Education

Diploma in Telecommunications Engineering

East African School of Aviation, Nairobi | Sep 2019 - Nov 2022

Kenya Certificate of Secondary Education

Midland Ridge High School, Nairobi | Feb 2015 - Dec 2018